AHMAD ZACKI HABIBI

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Electrical Engineering graduate (S.T.) from Telkom University certified in PLC & Industrial Automation, with a solid grounding in electrical power systems, circuit schematics, 3-phase diagnostics, and industrial equipment maintenance. Experienced in technical maintenance and troubleshooting on critical electronic systems and control panels at PT Angkasa Pura II. Demonstrated capability in electrical system simulation (PVsyst Single Line Diagrams), power fault classification, and industrial sensor-actuator interfacing. Highly adaptable, disciplined in technical documentation, and ready to execute engineering projects across head office and project site assignments. Proficient in English (EPRT 527).

EDUCATION

Sarjana Teknik Elektro (S.T.) / B.S. in Electrical Engineering • Telkom University 2022 – 2026

Bandung, Indonesia | Focus: Electrical Power Systems, Industrial Automation & Control

- Completed Bachelor of Electrical Engineering curriculum covering electrical machines, power distribution, single-line diagrams, control systems, and instrumentation.
- Final Project: IoT-Based Real-Time Logistics Tracking System with Route Evaluation Using Rule-Based Scoring.
- English Proficiency: Telkom University EPRT Score 527 (Professional Working Proficiency in writing and speaking).

CERTIFICATIONS

- Programmable Logic Controller (PLC) & Industrial Automation: Industrial control logic, ladder programming, sensor-actuator interfacing, motor/drive control concepts, and automated process control.
- English Proficiency Test (EPRT): Score 527 • Telkom University Language Center (LaC).

WORK & ELECTRICAL MAINTENANCE EXPERIENCE

Intern, SSES (Safety, Security, and Electronic System) • PT Angkasa Pura II Jul 2025 – Aug 2025

Jakarta, Indonesia

- Assisted SSES engineering unit in preventive maintenance, technical troubleshooting, and performance verification of critical electrical and electronic equipment (switchgears, control panels, X-ray body scanners, WTMD, HHMD).
- Participated in circuit-level diagnostics and component troubleshooting for electronic equipment and network switches to maintain high operational availability and safety compliance.
- Collaborated with maintenance engineers in executing scheduled maintenance routines and recording technical equipment status reports.

ORGANIZATIONAL & TECHNICAL LEADERSHIP

Laboratory Assistant & Practicum Staff • uPCI (Microprocessor & Interface Laboratory) 2024 – 2026

- Guided engineering students in microcontroller architectures, I/O peripheral interfacing, circuit wiring schematics, and hardware troubleshooting (Assembly, C/C++).
- Managed weekly laboratory sessions, inspected equipment readiness and power supplies, and prepared structured technical evaluation reports.

Vice Coordinator of Logistics & Equipment • QUOTE 2024 (KMTE Telkom University) 2025 – 2026

- Oversaw technical equipment logistics, power infrastructure distribution, and site layout readiness for a major departmental event, achieving zero operational downtime.
- Coordinated cross-functional teams in equipment procurement, facility permitting, timeline scheduling, and on-site technical resolution under tight deadlines.

ENGINEERING & TECHNICAL PROJECTS

Solar PV System Design & Simulation (Single Line Diagram & Energy Analysis) PVsyst

- Modeled and simulated 4.1 kWp residential on-grid and off-grid solar photovoltaic systems, evaluating equipment layout, inverter matching, and electrical balance of system.
- Constructed Single Line Diagram (SLD) configurations and component datasheets in PVsyst, achieving an optimum Performance Ratio of 0.897 and 5,174 kWh/year yield with microinverter architecture.

  Tech Stack: PVsyst

Electrical Fault Detection & Classification in 3-Phase Power Systems Machine Learning

- Developed machine learning models (Random Forest and 1D CNN) to classify 5 electrical transmission and distribution fault types (Normal, LG, LL, LLG, LLL) based on 3-phase current and voltage waveforms.
- Achieved ~83% classification accuracy; deployed an interactive web application via Gradio for real-time power network fault diagnostic visualization.

  Tech Stack: Python, TensorFlow, Scikit-learn, Gradio

Smart Coffee Dryer • Industrial Process & Temperature Control System IoT & Industrial Control

- Engineered an automated hybrid drying system integrating a 500 W PTC heating element, DS18B20 temperature sensors, and load cell telemetry with Arduino Mega & ESP32.
- Implemented closed-loop temperature control (45-50°C) and real-time weight tracking to precisely regulate drying kinetics until reaching the target ~12% moisture content.
  Tech Stack: Arduino Mega, ESP32, PTC Heater (500W), DS18B20, Load Cell, C++

IoT Real-Time Logistics Tracking System with Route Evaluation (Undergraduate Thesis) Tugas Akhir

- Engineered an IoT fleet monitoring telemetry system integrating ESP32, u-blox Neo-M8U GNSS, and SIM800L cellular module, streaming real-time GPS telemetry via MQTT to a FastAPI backend.
- Achieved 100% packet delivery rate at 213-215 ms average latency, 100% Haversine route deviation detection rate across 5 test locations, and <1s adaptive rerouting via OSRM.
  Tech Stack: ESP32, GNSS (u-blox Neo-M8U), SIM800L, MQTT, FastAPI, Leaflet.js, Python

TECHNICAL SKILLS & COMPETENCIES

Electrical & Industrial Automation: PLC (Programmable Logic Controller), Industrial Automation, 3-Phase Electrical Power Systems, Single Line Diagram (SLD) Design, Electrical Circuit Troubleshooting, Motor & VFD Concepts, Sensor-Actuator Interfacing, Control Panel Diagnostics, Electronic Equipment Maintenance (SSES), Multimeter & Oscilloscope.

Engineering Software & Tools: PVsyst, Fusion 360, Arduino IDE, Git, Google Colab, Linux/Windows.

Programming & IoT Protocols: Python, C/C++, Java, Assembly, MATLAB, SQL, MQTT, WebSocket, FastAPI.

Languages & Professional Skills: English (Telkom University EPRT Score: 527 • Professional Working Proficiency), Indonesian (Native), Technical Documentation, Project Scheduling & Logistics Management, Cross-Functional Team Collaboration.